

February 27, 2019



U.S. Department
of Transportation

East Building, PHH-30
1200 New Jersey Avenue S.E.
Washington, D.C. 20590

**Pipeline and Hazardous
Materials Safety Administration**

DOT-SP 15720
(THIRD REVISION)

EXPIRATION DATE: 2023-01-31

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Hexagon Digital Wave LLC
Englewood, CO
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the extension of the service life of certain composite cylinders owned by the US Navy which are subjected to requalification procedures and operational controls that are defined in this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.205(g) in that alternative testing is authorized.
5. BASIS: This special permit is based on the application of Hexagon Digital Wave LLC dated February 19, 2019 submitted in accordance with § 107.109.

Tracking Number: 2019025867

February 27, 20196. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed (breathing air containing up to 39% by volume oxygen)	2.2	UN1002	N/A

7. SAFETY CONTROL MEASURES:

a. **PACKAGING:** Packaging prescribed are limited to certain U.S. Navy owned, non-DOT specification composite cylinders made under DOT FRP-1 and DOT CFFC standards and manufactured under special permits DOT-SP 7277, DOT-SP 10915 and DOT-SP 10945, used exclusively in Self Contained Breathing Apparatus (SCBA) service and maintained under the Navy standard operating procedure as described in the Digital Wave application submitted to the Office of Hazardous Materials Safety Approvals and Permits Division (OHMSAPD). Cylinders may be considered for an extended service life not exceeding 30 years after the original manufacturing date of the cylinder. Each cylinder considered for extended service life must be requalified by Modal Acoustic Emission (MAE) in accordance with the procedures specified in the Digital Wave Corporation (DWC) application for special permit on file with OHMSAPD and as prescribed in this special permit. Cylinders passing the MAE requalification may be marked DOT-SP 15720 and authorized for an additional 5 years. Cylinders marked DOT-SP 15720 must be successfully requalified once every 5 years using MAE in order to remain in service for a maximum service life of 30 years from the date of original manufacture.

b. **MAE Equipment and Performance:** The MAE testing system must include 1) piezoelectric sensors, 2) preamplifiers, 3) high-pass and low-pass filters, 4) amplifiers, 5) A/D (analog-to-digital) converters, 6) a computer program for the collection of data, 7) computer and monitor for the display of data, and 8) a computer program for analysis of data. The MAE technician must be capable of examining the

February 27, 2019

waveforms (event by event) and the waveforms for each event must correspond precisely with the pressure and time data during the test. The MAE testing system must include a computer program capable of detecting the first arrival channel. The MAE testing system must include sensors and recording equipment with a current calibration sticker (yearly) or a current certificate of calibration. Sensors shall have a flat frequency response (+/- 6 dB amplitude response over the frequency range specified) from 50 kHz to 400 kHz. Deviation from flat response (signal coloration) must be corrected by using a sensitivity curve obtained with a Michelson Interferometer calibration system similar to the apparatus used by NIST (National Institute for Standards and Technology). MAE sensors must have a diameter no greater than 0.5 inch for the active part of the sensor face. The aperture effect must be taken into account. Sensor sensitivity must be at least 0.1 V/nm. Preamplifiers and amplifiers will have a flat frequency response (+/- 1 dB) over the sensor frequency specified. The MAE system must include a high-pass filter of 20 kHz. Also, a low-pass filter must be applied to prevent digital aliasing that occurs if frequencies higher than the Nyquist frequency (half the Sampling Rate) are in the signal. The MAE system must include the memory depth (wave window length) and sampling rate of the high speed analog-to-digital (A/D) converter and must be set for the MAE test requirements.

c. **Standard References and Calibration**

(1) The MAE system must be calibrated to detect and measure the wave energy of the test object (e.g. fiber breakage from a composite cylinder) by using a Rolling Ball Impactor and Inclined Plate. The rolling ball impactor must be used to create an acoustical impulse in the aluminum-inclined-plate. The impact setup includes a steel ball ½ inch in diameter. The ball impactor is made of chrome steel alloy hardened to R/C 63, ground and lapped to a surface finish of 1.5 micro-inch, within 0.0001 of actual size and roundness within 0.000025 inch. The calibration Inclined Plate is made of aluminum alloy 7075-T6, and must be at least 4' x 4' in size, and 0.125 inch (0.003 meters) in thickness and be supported by steel blocks. The inclined plate includes a machined square groove 3/8" wide which supports and guides the impact ball to the impact point. The length of groove and inclined angle must be 16" and 6° respectively. The grooved inclined plate

February 27, 2019

must be positioned next to the edge of the aluminum plate such that the center (equator) of the ball impacts the center (mid-line) of the edge of the aluminum plate, as shown in the figure 1 below.

(2) Each Sensor has a flat frequency response (+/- 6 Db amplitude response over the frequency range specified) and has a range of 50 kHz to 400 kHz and diameter less than or equal to 0.5 inch. Each sensor must be mounted on the inclined aluminum plate and tested separately via the rolling ball impact. The vertical position of the ball impact point must be adjusted gradually in order to "peak up" the acoustical signal, much as is done in ultrasonic testing where the angle is varied slightly to peak up the response. The center frequency of the first cycle of the extensional mode plate wave (E wave) must be confirmed as 125 kHz $\pm$ 10 kHz. The energy value in joules of the first half cycle of the E wave, is defined as U_{RBI}^{AE} , the wave energy detected by the MAE system, and must be used to scale the fiber break energy, U_{FB} . This must be an "end-to-end" calibration meaning that the energy measured using the complete AE instrumentation (sensor, cables, preamplifiers, amplifiers, filters and digitizer) that are to be used in the actual testing situation. The energy linearity of the complete AE instrumentation (sensor, cables, preamplifiers, amplifiers, filters and digitizer) must be measured by using different roll lengths of 8, 12 and 16 inches. A mechanism (manual or automated) must be used to release the ball down the inclined plane. The start of the E wave must be from the first cycle of the waveform recognizable as the front end of the E wave to the end of the E wave which taken as 10 μ s later (the time was calculated from the dispersion curves for the specified aluminum plate). The wave energy must be computed and recorded from the digital waveform as described in DWC application on file with OHMSAPD.

February 27, 2019

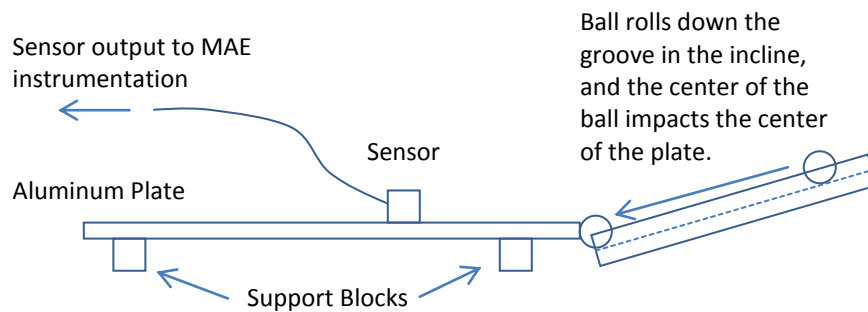


Figure 1. Inclined grooved Plate, impact balls, aluminum plate and sensor configuration

(3) Analog-to-Digital Converter Calibration - The sampling rate of the A/D must be such that aliasing does not occur. The recording system (consisting of all amplifiers, filters and digitizers beyond the sensor) must be calibrated by using a 20 cycle long tone burst with amplitude of 0.1 V at 100, 200, 300, and 400 kHz. The system must display an energy value of

$$U = (V^2 \times N \times T) / (2Z) \text{ joules}$$

at each frequency, where $V = 0.1$ volts, $N = 20$, Z is the preamplifier input impedance and T is the period of the cycle. If the measured values agree to within 15% of the theoretical values, the correct sampling rate has been set. These measurements must be performed at a system gain of 0 Db.

(4) Amplifier Gain Correction - All energies must be corrected for gain (e.g., 20 dB ($\times 10$) gain increases apparent energy 100 times and 40 dB ($\times 100$) gain increases apparent energy 10,000 times).

d. **Visual Inspection:** Prior to MAE testing, external and internal surfaces of each cylinder must be inspected in accordance with CGA Pamphlet C-6.2 Standards for Visual Inspection and Requalification of Fiber Reinforced High Pressure Cylinders, 1996, Third Edition. Cylinders that do not meet the visual inspection criteria must be condemned.

e. **MAE Test Procedure:** After completion of MAE system calibration, the following test procedure must be completed:

February 27, 2019

(1) Sensor Mounting - Two (2) sensors must be mounted on each cylinder, one sensor installed at each end of a cylinder. The sensors are located within two inches of the dome-to-sidewall transition area and will be in-line along the axial direction of the vessel.

(2) System sensitivity - The sampling rate and memory depth settings for the MAE system are:

(i) Threshold: 52 dBAE (The sensitivity must be adjusted to account for the response of the sensors as measured from the rolling ball calibration);

(ii) Sampling Rate: 5 MHz;

(iii) Memory Depth: 2048 Samples.

(3) Sensor Performance Checks - Hexagon Digital Wave LLC must conduct sensor performance checks prior to each test to verify proper operation and good coupling to the vessel. For the coupling check, the E and F waveforms shall be observed by breaking pencil lead at approximately 2 in. (5 cm) from each sensor along the axial direction of the vessel. The energy of the lead break waveforms shall be 5×10^{-15} to 20×10^{-15} Joules. If this energy level is not met, the sensor coupling shall be checked, or the sensor replaced. All calibration data shall be recorded. Amplitude response performance checks shall be carried out by pencil lead breaks (Pentel 2H, 0.3 mm) at a location centered along a line between the two sensors. Both sensors shall have a maximum amplitude response within 3 dB of each other. The gain settings for the calibration must be such that the signal does not saturate either the amplifiers or the A/D. If so, repeat the lead breaks at a system gain that does not saturate the system. Prior to pressurization, reset the gain to the test gain.

(4) Pressurization Procedure - Each cylinder must be subjected to a two-step hydraulic pressurization process from 0 psig to the cylinder's design test pressure (5/3 marked service pressure). During the first pressurization, the cylinder must be held at test pressure for at least 5 minutes and up to 15 minutes. If no MAE activity is recorded after a five (5) minute interval during the hold, the cylinder is stable and

February 27, 2019

the pressure may be reduced to zero (0) psig. Hold the pressure at zero psi for 100 seconds, and then repeat the step 1 pressurization, hold for a period of 100 seconds and depressurize as described above. Monitor and record the MAE event waveforms during the two-step pressurizations process. Pressurization must be stopped, if the Background Energy Oscillation (BEO) exceeds two (2) times the quiescent background energy of that channel. The fill rate must be less than the rate at which flow noise first appears. If at any time during fill the fill rate is too high in that it causes flow noise, decrease the fill rate until the flow noise disappears. Conduct a post-test system sensitivity check (lead breaks as described above) and save the data. The test temperature shall be between 50°F (10°C) and 120°F (49°C).

Warning: MAE testing must be done hydraulically and appropriate measures shall be taken to ensure safe operation and to contain any energy that may be released as a result of a cylinder rupture during pressurization.

(5) Stability Criteria During Pressure Holding - Upon filtration of unwanted data such as electromagnetic interference, flow noise and mechanical noise, the cumulative events versus time curve for each channel must be plotted. Also for each plot the event decay rate (B) as described in 7.f.(1), must be established.

In case of malfunction of the test equipment, the test may be repeated up to two times.

f. **Accept/Reject Criteria** - Each cylinder must be evaluated during MAE testing for the following four (4) Criteria:

(1) Stability - For each channel, the cumulative events versus time plot or event decay rate, B must be measured.

Rejection - a cylinder must be rejected if the value of B does not meet the following:

$$-0.1 \leq B \leq -0.0001, R^2 \geq 0.80;$$

B = event decay rate

February 27, 2019

R = coefficient of determination used in regression calculation.

Note: The above formula has been developed to meet the accept/reject criteria relevant to the fibers used to manufacture DOT CFFC and DOT FRP-1 cylinders.

(2) Fiber Breakage - To determine if fiber bundle breakage has occurred during the second pressurization, the frequency spectra of the direct E and F waves in the digital signal must be examined as following:

(i) Energy Measurement - The energies, U, for the following frequency ranges must be measured and recorded:

U0: 50 - 400 kHz
U1: 100 - 200 kHz
U2: 250 - 400 kHz

(ii) Energy Conditions- The criteria for determining if high frequency spectrum events have occurred, the following conditions must be calculated:

A - $U0 \geq U_{FBB}^{AE}$ Joules
and
B - $U2 / (U1 + U2) \geq 30\%$
and
C - $U2 / U0 \geq 30\%$

(iii) The values of U_{FBB}^{AE} for DOT CFFC (Carbon Fiber) and DOT FRP-1 Glass Fiber cylinders are:

U_{FBB}^{AE} for DOT CFFC = 2.7×10^{-16} Joules
 U_{FBB}^{AE} for DOT FRP-1 = 1.5×10^{-15} Joules

Note: These values are sensor and system specific, and must be determined for each monitoring system configuration.

Rejection - a cylinder must be rejected if all three Energy Conditions (**A**, **B** and **C**) described in paragraph 7.f.(2)(ii) above have occurred.

February 27, 2019

(3) Friction between Fracture Surfaces - The energy from friction between fracture surfaces is measured from the recorded MAE events.

Rejection - a cylinder must be rejected if a measured MAE event (energy) is greater than 2.7×10^{-14} Joules for carbon fiber and 1.5×10^{-13} Joules for glass fiber.

(4) Background Energy Oscillation (BEO) - the background energy oscillation of each MAE channel must be measured and recorded during pressurization.

Rejection - a cylinder must be rejected if the BEO exceeds two (2) times the quiescent background energy of that channel at any time during pressurization.

Warning - If BEO with a difference in amplitude is greater than two (2) between minima and maxima, the pressure of the cylinder must be reduced immediately.

g. **Action for Rejected Cylinders:** When a cylinder is rejected, the retester must securely affix to the cylinder a label with the word "CONDEMNED" overcoated with epoxy near, but not obscuring, the original cylinder manufacturer's label. Alternatively, at the direction of the owner, the requalifier may render the cylinder incapable of holding pressure.

h. **Marking:** Each cylinder that has passed the requalification described in this special permit for extended service life, must be marked "DOT-SP 15720". The new "DOT-SP 15720" marking must cover the current special permit marking (DOT-SP 7277, DOT-SP 10915, or DOT-SP 10945) and then be epoxy coated to ensure it is permanently attached to the cylinder. The DOT-SP 7277, DOT-SP 10915, or DOT-SP 10945 marking and all indications of the identity of the party that manufactured the cylinder under DOT-SP 7277, DOT-SP 10915, or DOT-SP 10945 must be completely obscured. All marking, including requalification date (month/year), must be permanently placed on the cylinder as specified in paragraph § 180.213. In addition, each cylinder must be marked "MAE", in characters not less than 1/4". The marking of the RIN number on the cylinder certifies compliance with all of the terms and conditions of this special permit.

i. **Report:** A report must be generated for each cylinder that is tested. The MAE reports must include the following:

February 27, 2019

- (1) RIN
- (2) MAE equipment, model and serial number
- (3) Cylinder manufacture date, serial number, special permit number (DOT-SP 7277, DOT-SP 10915, DOT-SP 10945 or DOT-SP 15720) and marked service pressure
- (4) MAE technicians' name and certification level
- (5) Test date
- (6) Decay rate B values
- (7) Event energies exceeding the fiber bundle failure criteria, if any
- (8) Background energy oscillation pressure
- (9) Visual inspection results (Pass/Fail)
- (10) Acceptance/rejection results. If cylinder is rejected, include each cause of rejection criteria as described in 7.f.

The MAE reports must be on file at the test site, and made available to a DOT representative when requested.

j. **Personnel Qualification:** Each person who performs retesting or who evaluates or certifies retest results must meet the following requirements:

(1) Project Manager - is the senior manager of Hexagon Digital Wave LLC responsible for compliance with DOT regulations including this special permit. Additionally, the project manager must ensure that each operator and senior review technologist maintains the required certifications described herein.

(2) The personnel responsible for performing cylinder retesting under this special permit shall be qualified to an appropriate Level (Level I, II or III) in MAE testing in accordance with the American Society for Nondestructive Testing (ASNT) Recommended Practice SNT-TC-1A-1996 depending upon the assigned responsibility as described below:

(i) As a minimum, a Level II Operator must perform system startup, calibrate the system, and review and certify the test results when written acceptance and rejection criteria for cylinders has been provided by a Senior Review Technologist. Based on written criteria, the Level II Operator may authorize cylinders that pass the retest to be marked in accordance with paragraph 7.h. of this special permit. However, a person with Level I

February 27, 2019

certification may perform a system startup, check calibration, and perform MAE under the direct guidance and supervision of a Senior Review Technologist or a Level II Operator, either of whom must be physically present at the test site so as to be able to observe examinations conducted under this special permit.

(ii) Senior Review Technologist (SRT) - is a person who provides the written MAE procedure, supervisory training and examinations (Level I and II), technical guidance to operators and reviews and verifies the test results. An SRT must have a thorough understanding of the DOT Regulations (49 CFR) pertaining to the requalification and reuse of DOT cylinders authorized under this special permit and ASNT Recommended Practice SNT-TC-1A and possess either:

(A) A Level III certificate from ASNT in Acoustic Emission, with a documented minimum of 2 years of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines using the Modal Acoustic Emission test method;

(B) A Professional Engineer (PE) License with a documented minimum of 2 years of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines using the Modal Acoustic Emission test method; or,

(C) A PhD degree in Engineering/Physics with documented evidence of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines, using the Modal Acoustic Emission test method or research/thesis work and authoring/co-authoring of technical papers published, in recognized technical journals, in the field of Modal Acoustic Emission testing methods.

(D) The SRT must prepare and submit the reports required in paragraph 7.i. and annually verify that the MAE program is being operated in accordance with the requirements of this special permit.

February 27, 2019

The most recent copies of certification (e.g. ASNT Level III or PE license) must be available for inspection at each requalification facility.

k. **OPERATIONAL CONTROLS:**

(1) The cylinders that are requalified for service life extension under this special permit are limited to US Navy-owned cylinders, used exclusively in SCBA service and maintained under the Navy's standard operating procedures as described in the Digital Wave application submitted to OHMSAPD;

(2) Each cylinder that is requalified for service life extension under this special permit must be identified by special permit (DOT-SP 7277, DOT-SP 10915, DOT-SP 10945 or DOT-SP 15720), serial number, DOT standard (e.g. FRP-1 or CFFC), cylinder owner (Navy) and MAE requalification results (fail/pass);

(3) The original special permit marking (DOT-SP 7277, DOT-SP 10915, or DOT-SP 10945) of each cylinder that is requalified for service life extension under this special permit must be covered and replaced with DOT-SP 15720 as described in paragraph 7.h. of this special permit;

(4) Record Keeping - Prior to marking a cylinder with "DOT-SP 15720" (designation for extended service life), the identification of each cylinder as described in paragraph 7.k.(2) of this special permit must be recorded and kept of file. The record must be made available upon request;

(5) Revalidation Plan- All cylinders marked "DOT-SP 15720" are subject to the in-service life extension (Revalidation Plan). The Revalidation Plan is described in paragraph 8.a. of this special permit;

(6) Prior to filling, each cylinder marked "DOT-SP 15720" must be inspected visually for external damage (e.g. acid attack, fiber cut, surface impact, etc.). A cylinder with external damage must be further evaluated in accordance with CGA Pamphlet C-6.2. Cylinders that do not meet acceptance criteria of CGA C-6.2 must be condemned; and

February 27, 2019

(7) Maximum Service Life - Cylinders requalified under this special permit are authorized for an additional 5 year service life extension. The maximum service life of any cylinder that was qualified under this special permit is 30 years from the date of original manufacture.

8. **SPECIAL PROVISIONS:**

a. **Revalidation Plan:** Hexagon Digital Wave LLC must formulate a plan for all the US Navy-owned cylinders that are marked "DOT-SP 15720" for the additional service life extension. The plan must be submitted to OHMSAPD within 12 months of the issuance of this special permit the report should be submitted to specialpermits@dot.gov and must include:

(1) A detailed procedure for obtaining the Navy-owned cylinders from the field and design qualification testing of each design type;

(2) A procedure for the random selection of a minimum of 5 cylinders of each design type (FRP-1, CFFC) which have been in service for 18, 21, 24 and 27 years;

(3) All randomly selected cylinders from the field must be tested in accordance with Sections 8.5.4, 8.5.5, 8.5.7 and 8.5.8 of ISO 11119-2:2002. Acceptance criteria must be as defined in ISO 11119-2:2002. For cylinders manufactured using glass fiber, residual stress measurements must be performed on a single cylinder to demonstrate whether the stress performance is below that required to initiate stress rupture;

(4) The complete MAE test report, including the original test data, must be submitted to the Associate Administrator for Hazardous Materials Safety for assessment within 30 days of completion of the test. Failure to meet the acceptance criteria specified in this special permit may result in the design being restricted to a maximum life of 15 years.

b. The designated Senior Review Technologist (SRT) must, within twelve months from the date the special permit is issued and annually thereafter, review the MAE program. The designated SRT must submit a letter to OHMSAPD verifying that the MAE program is meeting the terms and requirements of this special permit. The letter may be submitted with

February 27, 2019

the report required below. If submitted electronically, the letter should be submitted to specialpermits@dot.gov.

c. The total number of cylinders tested under this special permit must be reported by special permit number, type (e.g. CFFC), serial number and age. These results must be summarized and reported to DOT on an annual basis. A summary of the test results at each facility must be reported (electronic or paper) to the Associate Administrator for Hazardous Materials Safety annually to assess the effectiveness of the test program. If submitted electronically, the report should be submitted to specialpermits@dot.gov. The summary must include the total number of cylinders tested under this special permit grouped by type or special permit number, size and age. The summary must include the number of cylinders accepted, rejected, or condemned. For any rejected or condemned cylinder, the defect causing the rejection/condemnation must be fully characterized and profiled (i.e., cuts, abrasions, impacts, fiber failures, chemical damage, heat damage, etc.) and the specific size of the defect should be determined (i.e., length, depth, width, etc.). The cylinders that were condemned at visual inspection (prior to MAE) must also be included in the report.

d. A statement of qualifications and supporting documentation, for each "qualified MAE tester" used under this special permit and information in support thereof, must be maintained by Hexagon Digital Wave LLC. The statement for each "qualified MAE tester", must be made available upon request.

e. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

f. A current copy of the DOT special permit for the cylinders listed in paragraph 7.a. of this special permit must be maintained at each facility where retesting is performed.

g. A person who is not a holder of this special permit who receives a packaging covered by this special permit may reoffer it for transportation provided no modification or change is made to the packaging and it is reoffered for transportation in conformance with this special permit and the HMR.

February 27, 2019

- h. All cylinders that are tested, marked, maintained, or used under this special permit, cease to be considered DOT-SP 7277, DOT-SP 10915, and DOT-SP 10945 cylinders and Hexagon Digital Wave LLC assumes all responsibilities and liabilities as the "manufacturer" of these cylinders, as that term is defined and used in the Federal hazardous materials transportation law (49 U.S.C. §§ 5101-5128) and any and all regulations promulgated by PHMSA (49 CFR Parts 105-180).
- i. Hexagon Digital Wave LLC must hold a valid RIN approval that indicates authorization to conduct testing under the terms of this special permit.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, passenger vessel, and cargo aircraft only.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft, or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

February 27, 2019

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) – “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 – Immediate notice of certain hazardous materials incidents, and 171.16 – Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit; notification should be provided to specialpermits@dot.gov.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-30, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

February 27, 2019

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Andrew Eckenrode